Additional 2:
Design problem: Image you have a macadamia nut that you want to crack open by hand using a simple lever nut cracker.

a) Draw a figure of the nut cracker with the nut when it's about to crack. Calculate the necessary dimensions of the nutcracker and come up with a design to make this task feasible. You can assume a very simple geometry for the nutcracker to make your calculations easier.
   i. You will need to find typical values for the inputs (e.g. size for macadamia nuts and maximum grip strength).
   ii. The average load required to crack a macadamia nut can be found in the appendix section of this paper (Schrauf et al. Do capuchin monkeys use weight to select hammer tools, Anim Cogn 11, 413–422 (2008). https://doi.org/10.1007/s10071-007-0131-2)

b) Discuss the usability of the nutcracker that you designed

c) Now, instead of relying on grip strength to apply the input force, modify your design to use a linear actuator.
   i. Pick a linear actuator from this website, taking into account both force output and size/stroke.
   https://www.progressiveautomations.com/collections/linear-actuators

Given- Simple geometry
values guessed/found below

find- appropriate dimensions to crack the nut with appropriate size of nut and grip strength

Plan-
find appropriate values for size of nut and grip strength
Decide on a geometry
explode FBD halves and draw out forces
run MA analysis with $\sum M_A$ to find length of force applied to pivot-lever arm.

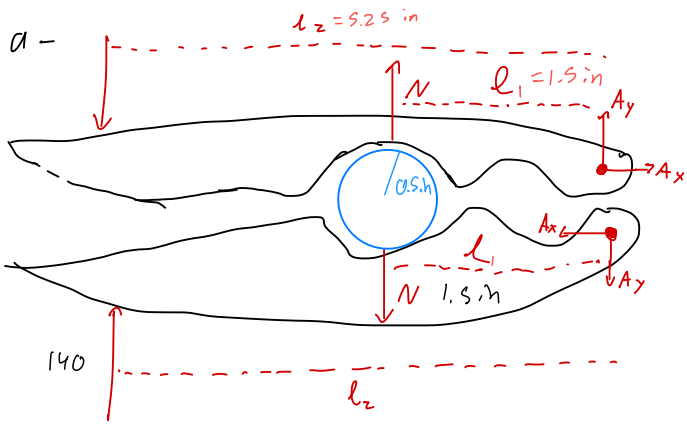
Using my hand dynamometer I bought for a school project in high school, I got 138.7 lb which I will approximate as 140 lb.

Averge diameter of a macadamia nut - 0.8-1.2 in, so I'll go right down the middle and use 1 inch, modeling it as a perfect sphere

Nut hardness → 222.18 kg on average.

↳ 222.18 kg ≈ 490 lb

a-



$$MA \text{ needed} = \frac{490}{140} = 3.5$$

Choose $l_1 = 1.5$ in arbitrarily, allows nut to enter

$$\sum M_A: 140 \cdot l_2 - 490(1.5) = 0$$

$$l_2 = 5.25 \text{ in}$$

b- 5.25 inches works well if you have a grip strength of 140 lb, which most people don't, and its a normal nut, it could be harder.
With the current specifications, we would want a much longer handle to make it easier/higher mechanical advantage. 8 inches would give a MA of $\frac{8}{1.5} = 5.33 > 3.5$

C- IPGS Mini Linear Actuator → 225 lb option, 2 in stroke

$F_{ACT} \cdot l_2 = 490 \cdot 1.5$ → $225 \cdot l_2 = 490 \cdot 1.5$ $l_2 = 3.27$ in from pin

2 in stroke allows jaws to close fully, 225 lb force allows the action to be closer

reflection- I think I did this right, directions weren't super clear.
Calculations were pretty basic compared to others, also not clear on what a linear actuators.